Neural Scaling in Robot Swarms

No Author Given

No Institute Given

A Appendix

Full Experimental Figures and Supplementary Results
This appendix contains all supplementary plots produced in the experiment. These include full average fitness curves, maximum-fitness plots, pairwise t-test heatmaps, genome-size analyses, and transfer-learning outputs. Only summary figures remain in the main text; all detailed plots are provided here for completeness and reproducibility.

Figures 1 and 2–4 present the genome size analysis and transfer-learning results, respectively. Additional maximum fitness plots and statistical heatmaps are provided in Figures 17–28 and Figures 29–31.

A.1 Average-Fitness Plots (All Variants)

Figures 5–8 show the average-fitness curves for all controller architectures in the Easy maze. Figures 9–12 show the corresponding results for the Medium maze, and Figures 13–16 present the results for the Difficult maze.

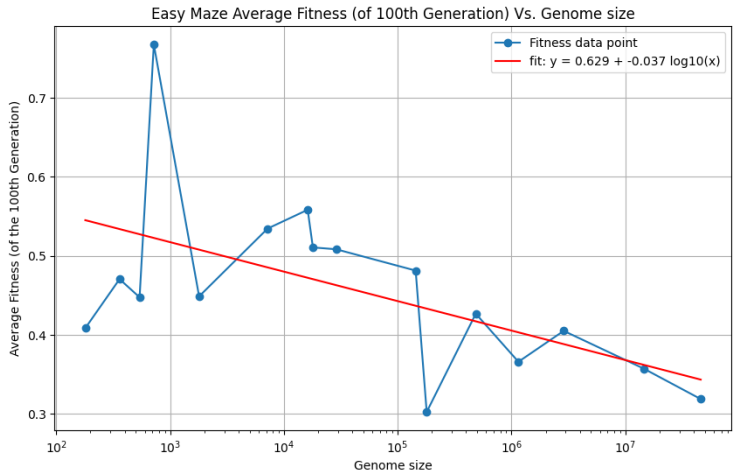
B Transfer Learning Results

This section presents the complete transfer-learning results used to compute the summary bar plots shown in the main text. Each table corresponds to a fixed *test environment*, and each row indicates the environment on which the controller was originally trained. Each column denotes a specific ANN variant.

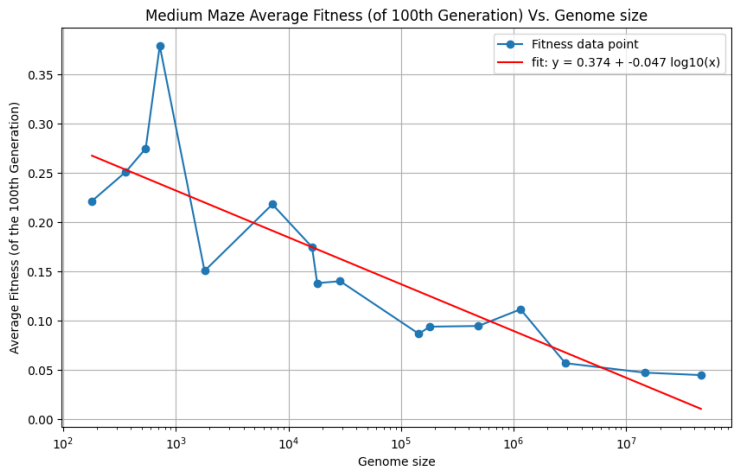
Entry(i,j): average fitness when variant j was trained on maze i and evaluated on the target maze.

Table 1: Transfer-learning results when testing on the Easy maze.

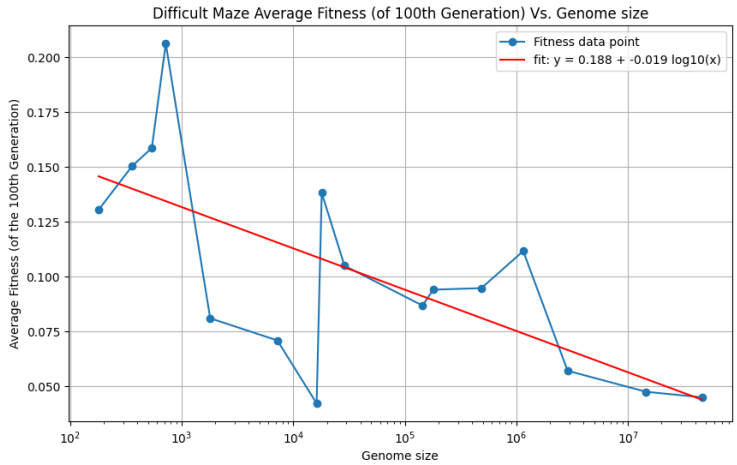
Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.465	0.575	0.190	0.695	0.015	0.040	0.005	0.055	0.015	0.085	0.045	0.010	0.025	0.025	0.025	0.085
Medium	0.090	0.625	0.195	0.885	0.065	0.030	0.040	0.015	0.075	0.025	0.035	0.010	0.090	0.005	0.000	0.055
Difficult	0.100	0.085	0.193	0.005	0.070	0.010	0.005	0.025	0.015	0.015	0.055	0.025	0.045	0.035	0.025	0.020



(a) Easy maze. Linear fit: $y = 0.629 - 0.037 \log_{10}(x)$.



(b) Medium maze. Linear fit: $y = 0.374 - 0.047 \log_{10}(x)$.



(c) Difficult maze. Linear fit: $y = 0.188 - 0.019 \log_{10}(x)$.

Fig. 1: Average fitness (at generation 100) vs. genome size ($\log_{10}$ scale). Blue: data; red: least-squares fit in $\log_{10}$-space.

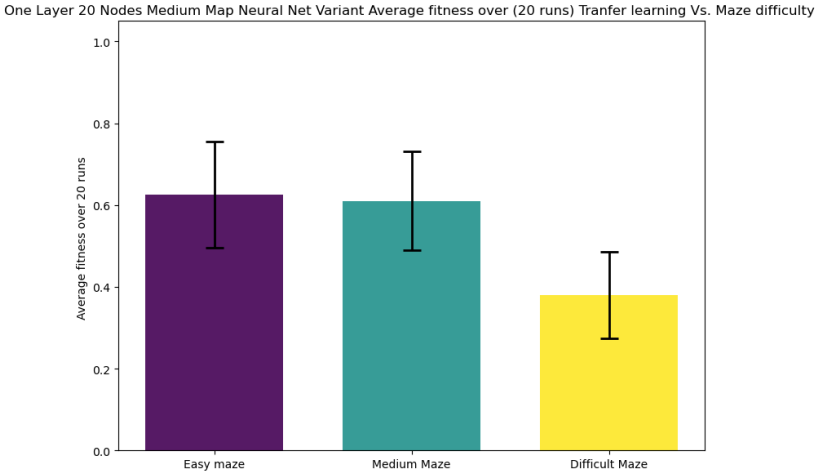


Fig. 2: Transfer-learning results for 1L20n controller.

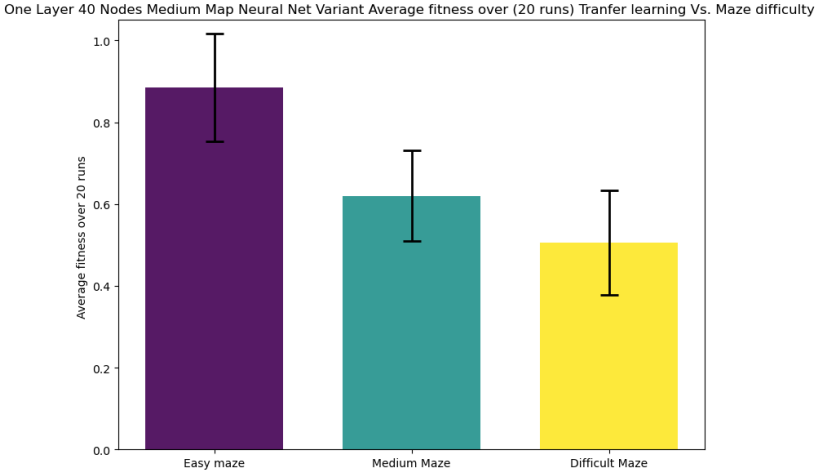


Fig. 3: Transfer-learning results for 1L40n controller.

Table 2: Standard deviations (over 20 runs) of transfer-learning fitness when testing on the Easy maze.

Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.172520	0.158529	0.137267	0.157196	0.048936	0.050262	0.022361	0.060481	0.036635	0.074516	0.082558	0.030779	0.055012	0.055012	0.063867	0.103999
Medium	0.102084	0.129269	0.146808	0.130888	0.108942	0.057124	0.059824	0.036635	0.085070	0.055012	0.058714	0.030779	0.085224	0.022361	0.000000	0.075915
Difficult	0.107606	0.098809	0.147800	0.022361	0.086450	0.030779	0.022361	0.055012	0.048936	0.048936	0.088704	0.063867	0.068633	0.048936	0.055012	0.052315

Table 3: Transfer-learning results when testing on the Medium maze.

Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.070	0.275	0.040	0.210	0.015	0.040	0.025	0.025	0.010	0.040	0.025	0.015	0.030	0.015	0.020	0.040
Medium	0.015	0.610	0.125	0.620	0.040	0.040	0.010	0.015	0.060	0.015	0.030	0.025	0.040	0.025	0.010	0.000
Difficult	0.040	0.180	0.135	0.050	0.055	0.015	0.025	0.020	0.020	0.035	0.045	0.030	0.050	0.040	0.010	0.030

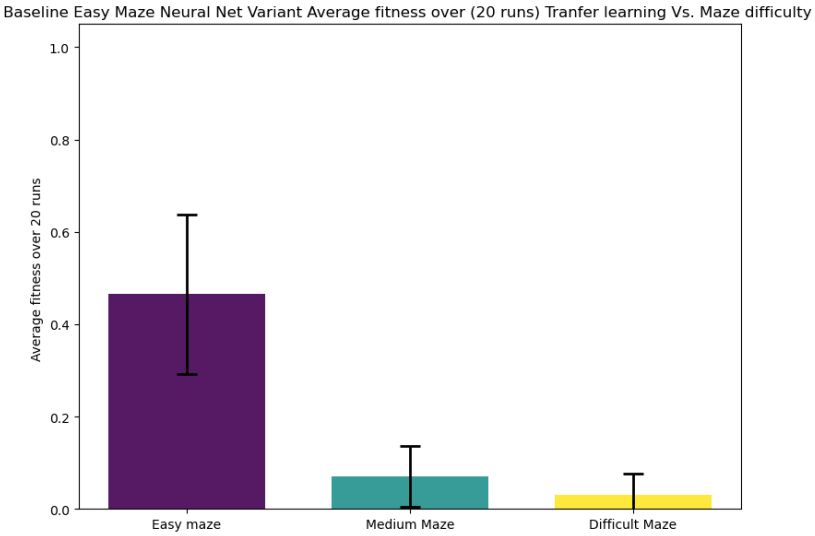


Fig. 4: Transfer-learning results for 1L10n (baseline) controller.

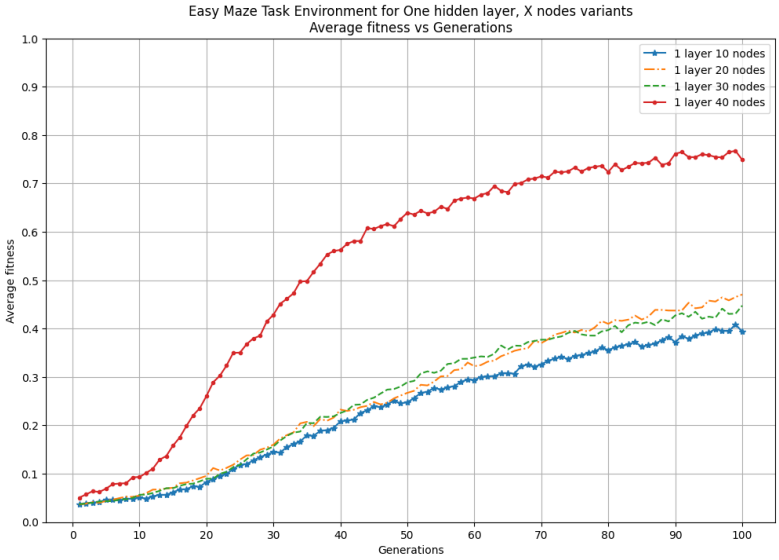


Fig. 5: Easy maze: Average fitness, one hidden layer.

Table 4: Standard deviations (over 20 runs) of transfer-learning fitness when testing on the Medium maze.

Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.065695	0.161815	0.059824	0.111921	0.048936	0.075394	0.055012	0.055012	0.030779	0.059824	0.044426	0.036635	0.073270	0.036635	0.041039	0.059824
Medium	0.036635	0.120961	0.125132	0.110501	0.050262	0.068056	0.030779	0.036635	0.068056	0.036635	0.057124	0.055012	0.059824	0.044426	0.030779	0.000000
Difficult	0.059824	0.139925	0.14623	0.082717	0.060481	0.036635	0.044426	0.041039	0.041039	0.067082	0.060481	0.073270	0.094591	0.059824	0.030779	0.065695

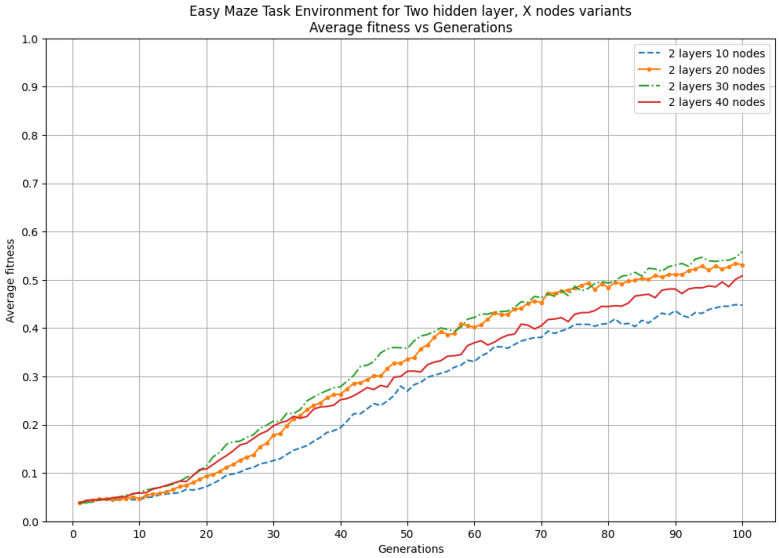


Fig. 6: Easy maze: Average fitness, two hidden layers.

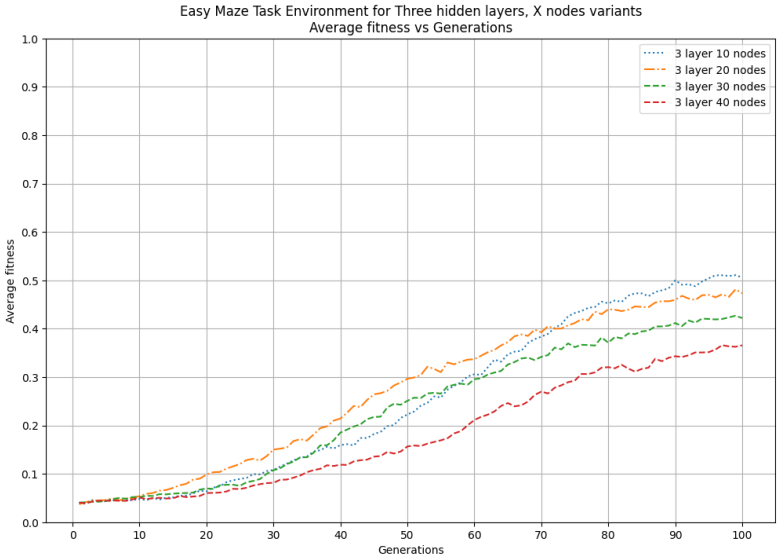


Fig. 7: Easy maze: Average fitness, three hidden layers.

Table 5: Transfer-learning results when testing on the Difficult maze.

Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.030	0.175	0.045	0.090	0.015	0.035	0.005	0.035	0.010	0.050	0.030	0.010	0.005	0.015	0.005	0.025
Medium	0.060	0.380	0.060	0.505	0.030	0.070	0.015	0.020	0.010	0.020	0.040	0.015	0.030	0.050	0.015	0.005
Difficult	0.010	0.135	0.033	0.040	0.020	0.025	0.010	0.040	0.015	0.025	0.025	0.035	0.040	0.020	0.010	0.015

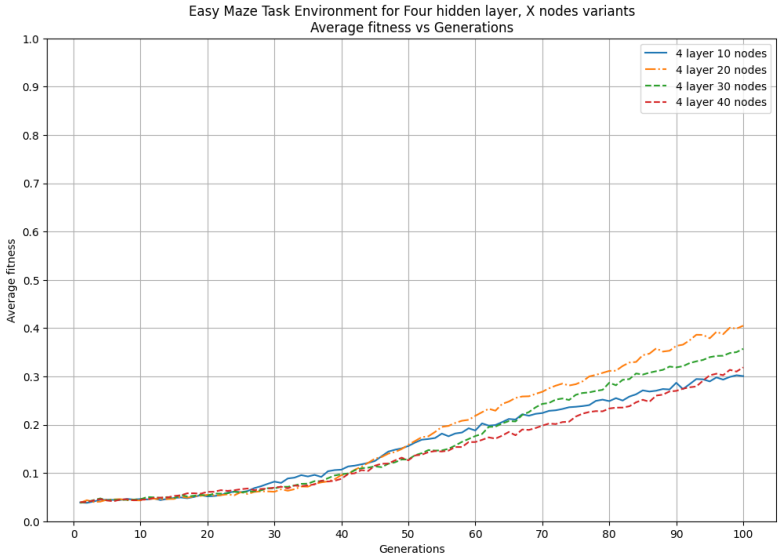


Fig. 8: Easy maze: Average fitness, four hidden layers.

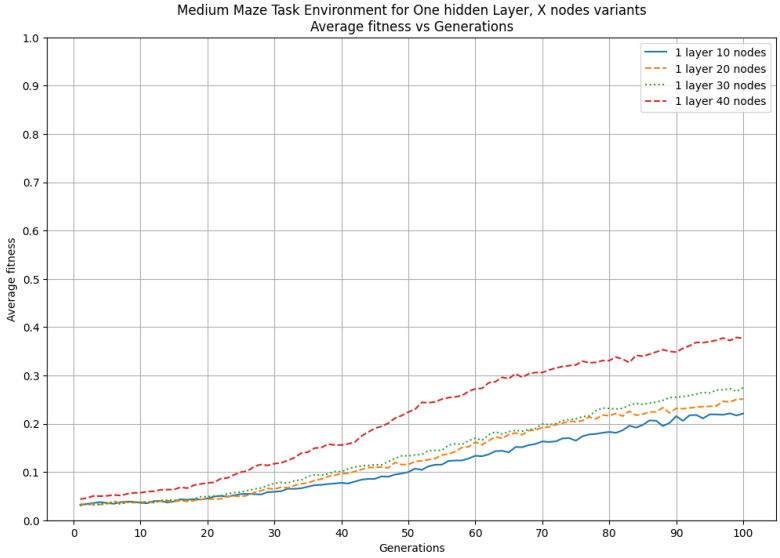


Fig. 9: Medium maze: Average fitness, one hidden layer.

Table 6: Standard deviations (over 20 runs) of transfer-learning fitness when testing on the Difficult maze.

Trained on	1L10n	1L20n	1L30n	1L40n	2L10n	2L20n	2L30n	2L40n	3L10n	3L20n	3L30n	3L40n	4L10n	4L20n	4L30n	4L40n
Easy	0.047016	0.101955	0.068633	0.071818	0.036635	0.048936	0.022361	0.074516	0.030779	0.068825	0.047016	0.030779	0.022361	0.036635	0.022361	0.055012
Medium	0.082078	0.105631	0.104630	0.127630	0.065695	0.065695	0.036635	0.052315	0.030779	0.052315	0.075394	0.036635	0.057124	0.082717	0.036635	0.022361
Difficult	0.030779	0.126803	0.034	0.050262	0.041039	0.055012	0.030779	0.068056	0.036635	0.063867	0.044426	0.067082	0.068056	0.052315	0.030779	0.036635

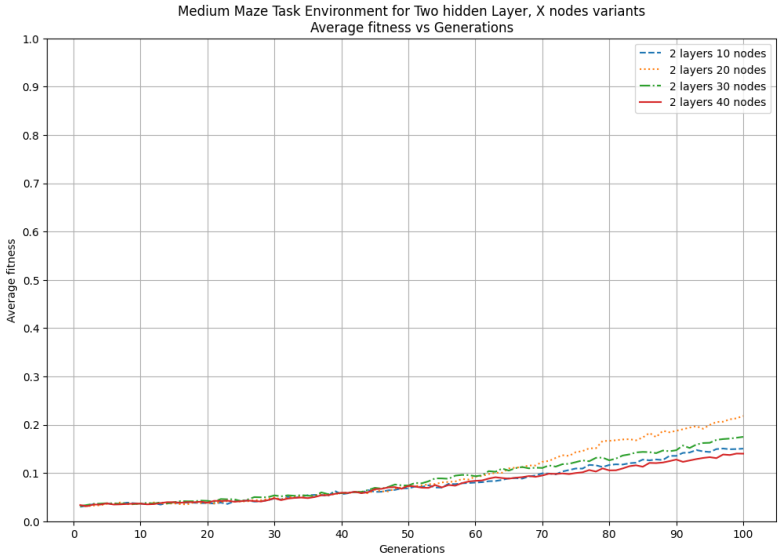


Fig. 10: Medium maze: Average fitness, two hidden layers.

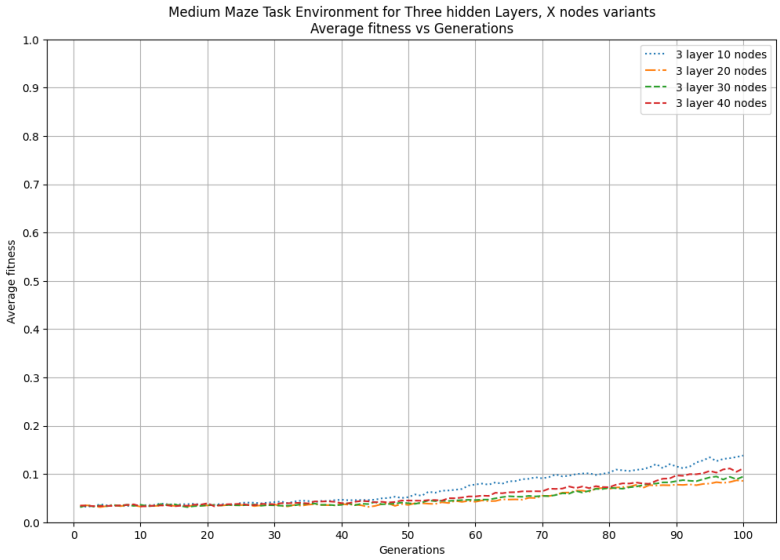


Fig. 11: Medium maze: Average fitness, three hidden layers.

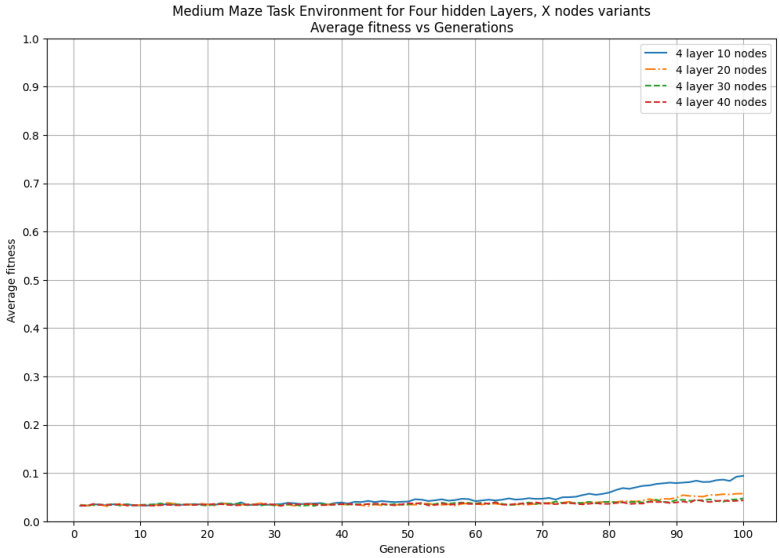


Fig. 12: Medium maze: Average fitness, four hidden layers.

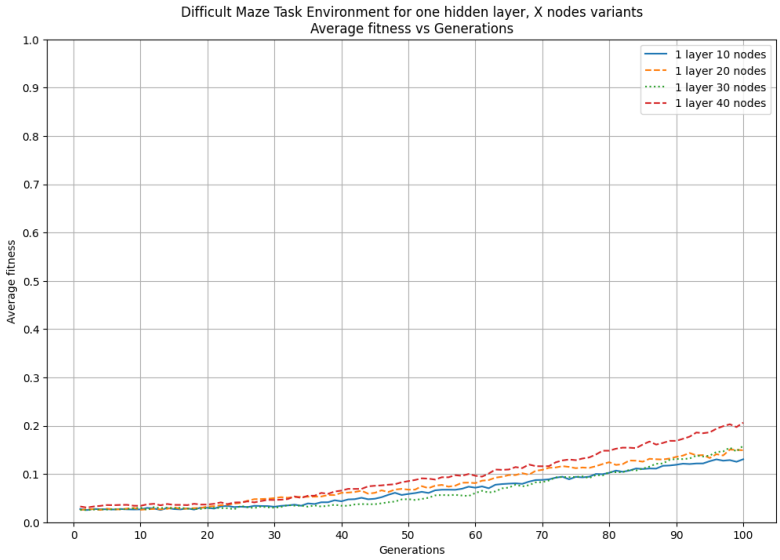


Fig. 13: Difficult maze: Average fitness, one hidden layer.

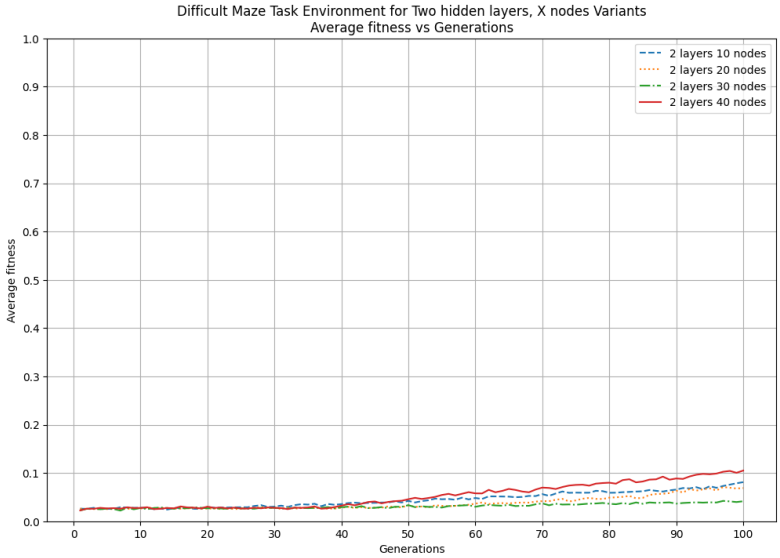


Fig. 14: Difficult maze: Average fitness, two hidden layers.

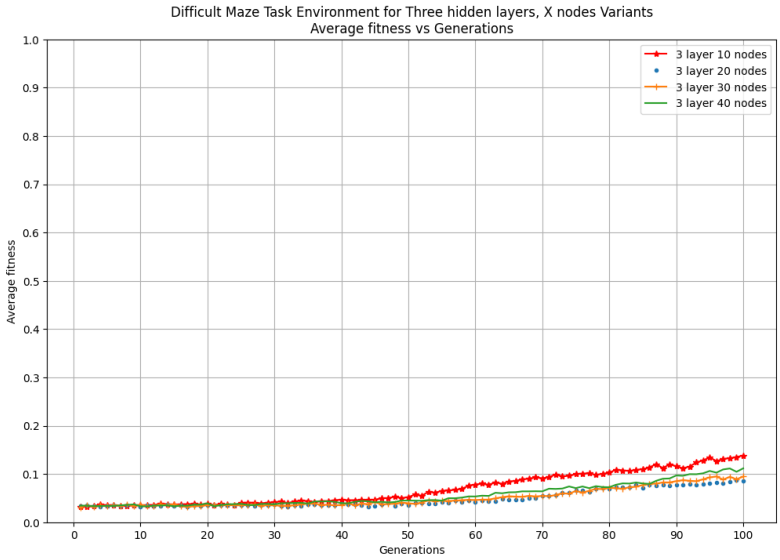


Fig. 15: Difficult maze: Average fitness, three hidden layers.

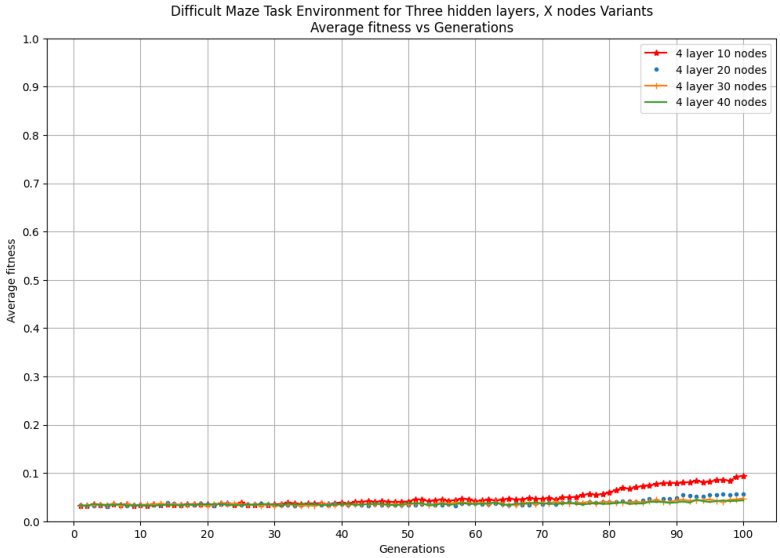


Fig. 16: Difficult maze: Average fitness, four hidden layers.

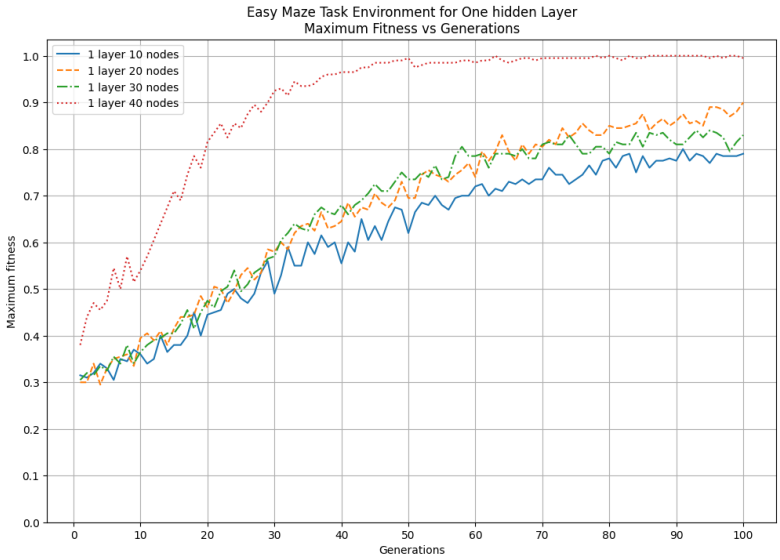


Fig. 17: Easy maze: Maximum fitness, one hidden layer.

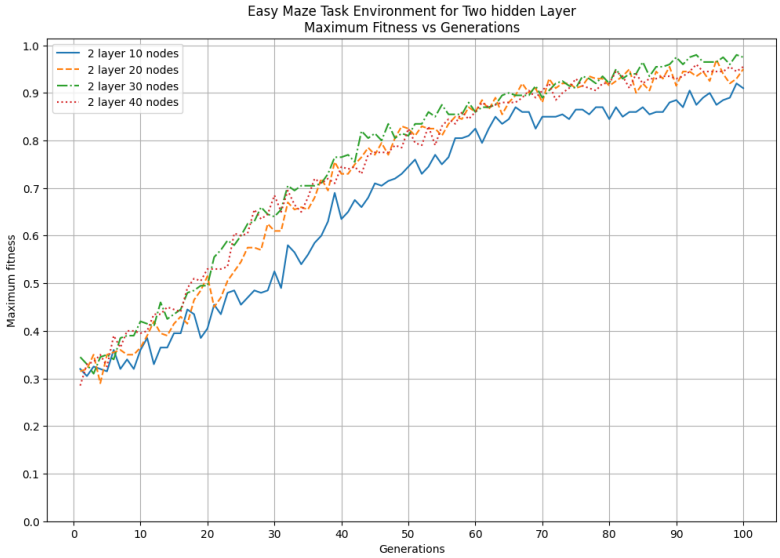


Fig. 18: Easy maze: Maximum fitness, two hidden layers.

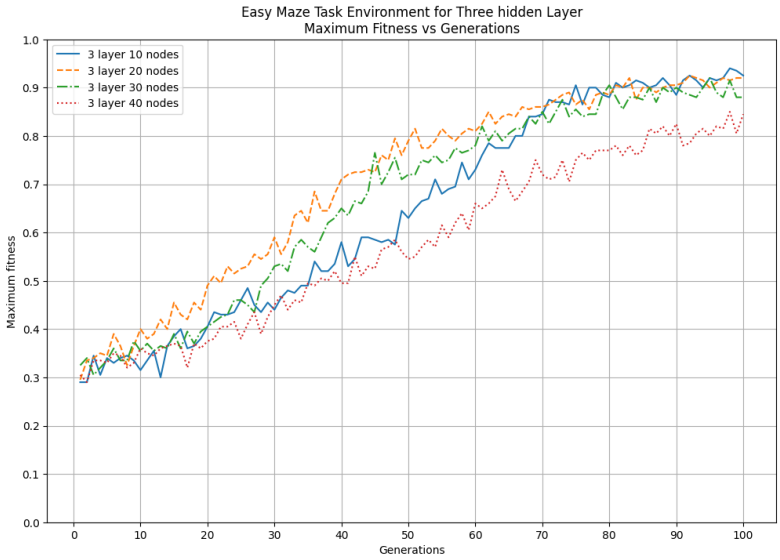


Fig. 19: Easy maze: Maximum fitness, three hidden layers.

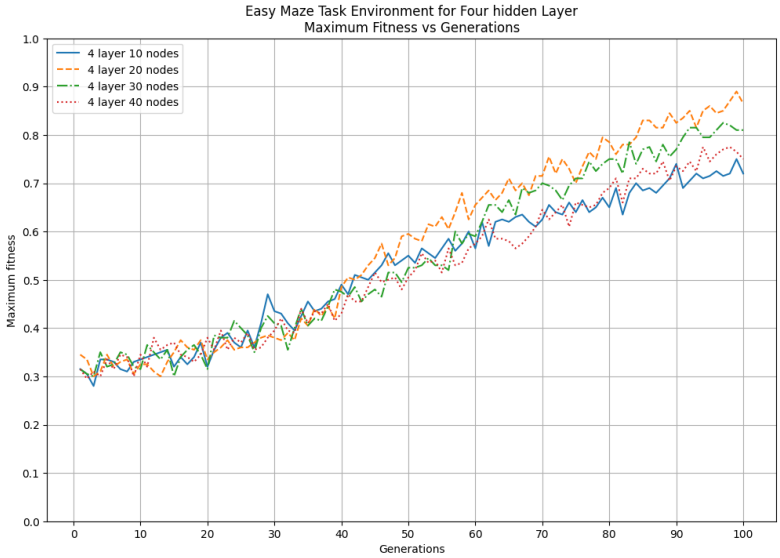


Fig. 20: Easy maze: Maximum fitness, four hidden layers.

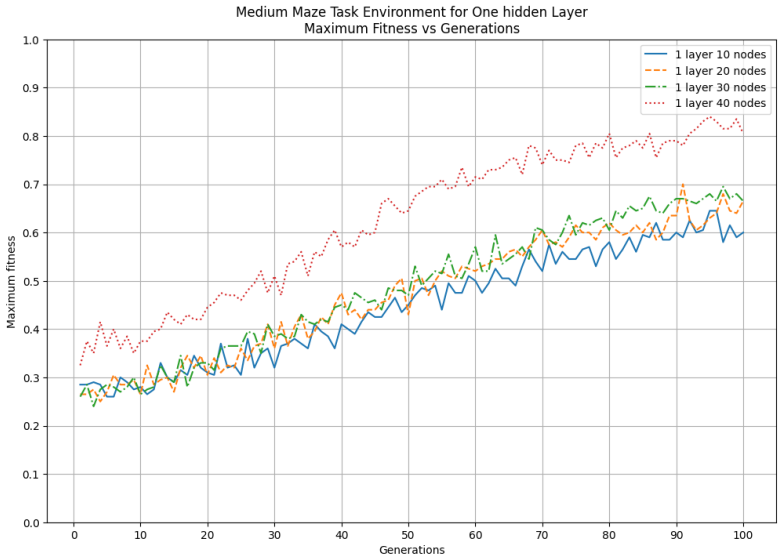


Fig. 21: Medium maze: Maximum fitness, one hidden layer.

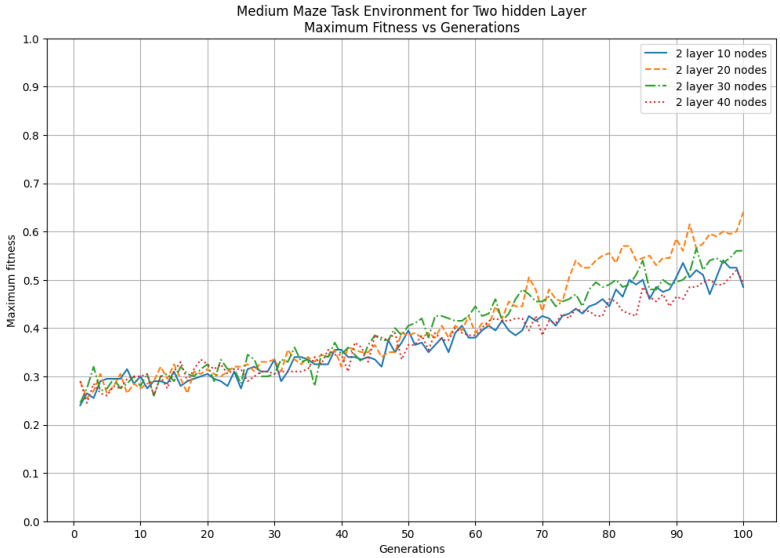


Fig. 22: Medium maze: Maximum fitness, two hidden layers.

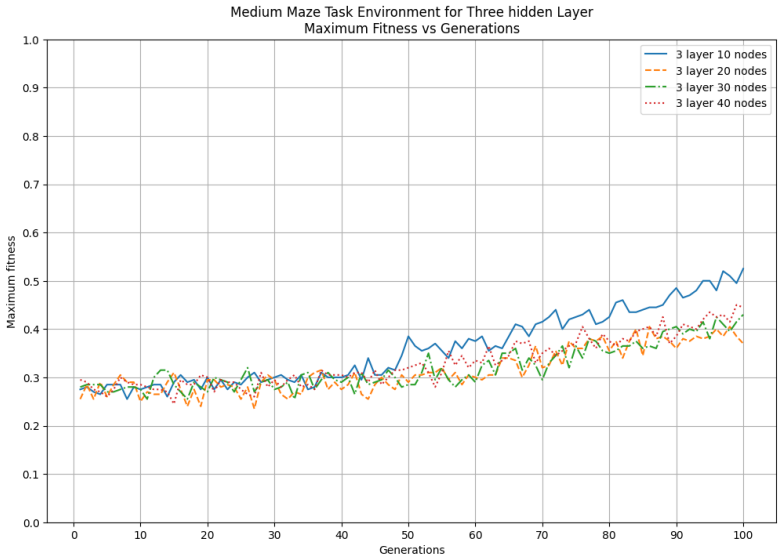


Fig. 23: Medium maze: Maximum fitness, three hidden layers.

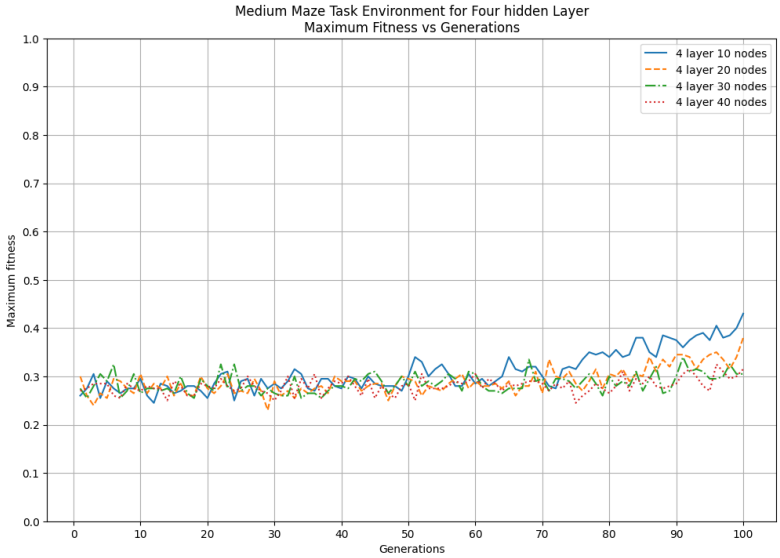


Fig. 24: Medium maze: Maximum fitness, four hidden layers.

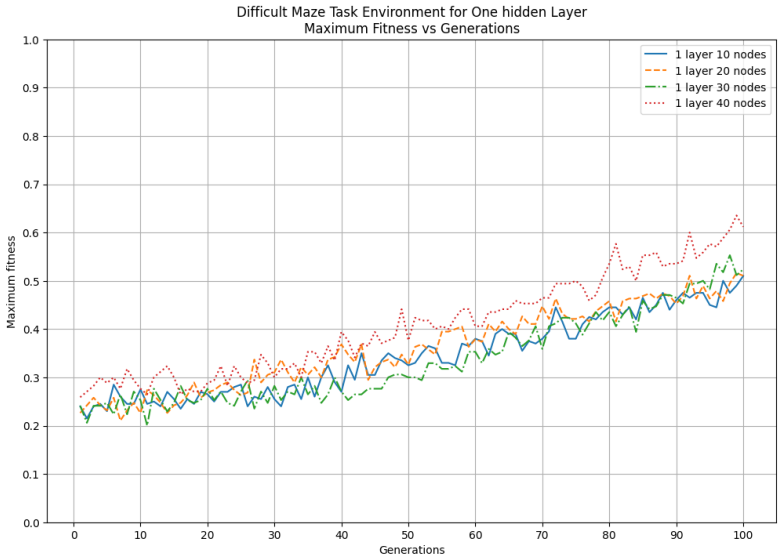


Fig. 25: Difficult maze: Maximum fitness, one hidden layer.

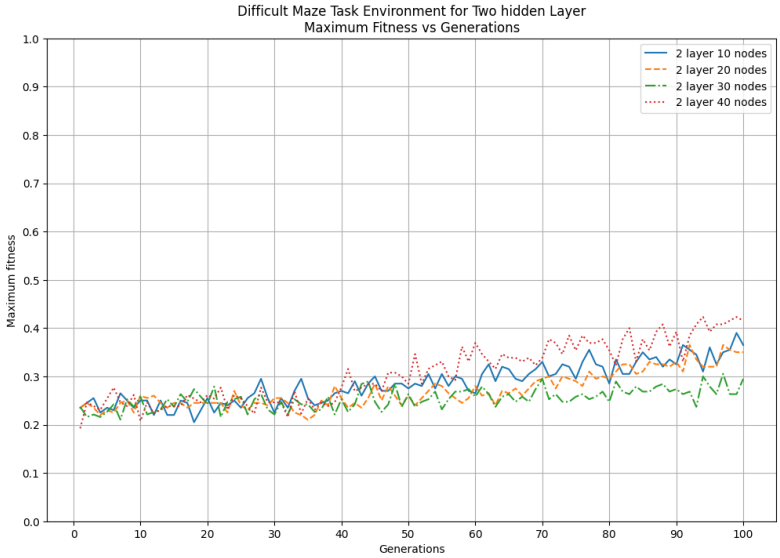


Fig. 26: Difficult maze: Maximum fitness, two hidden layers.

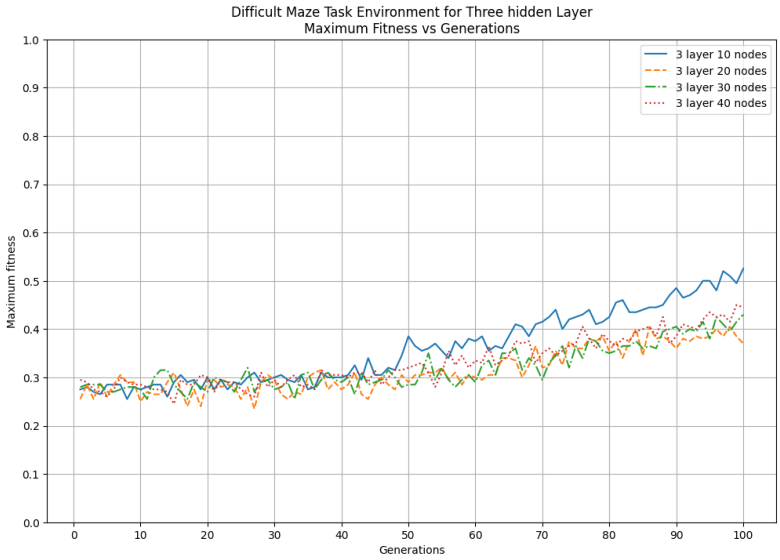


Fig. 27: Difficult maze: Maximum fitness, three hidden layers.

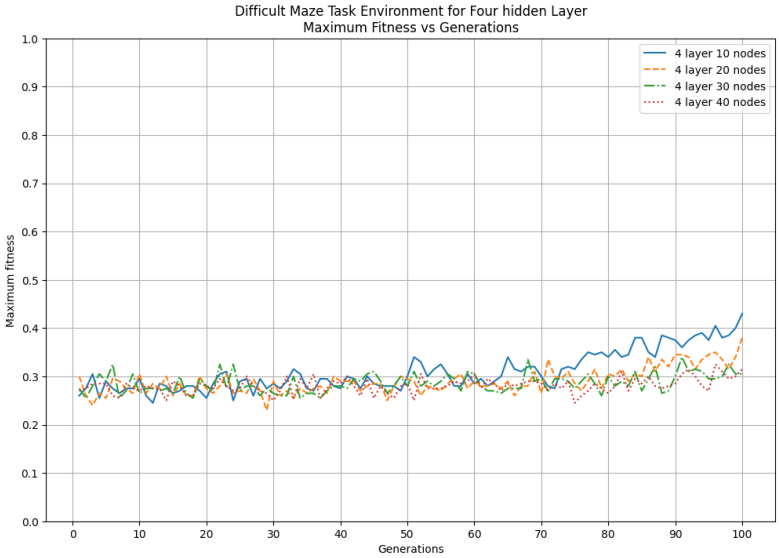


Fig. 28: Difficult maze: Maximum fitness, four hidden layers.

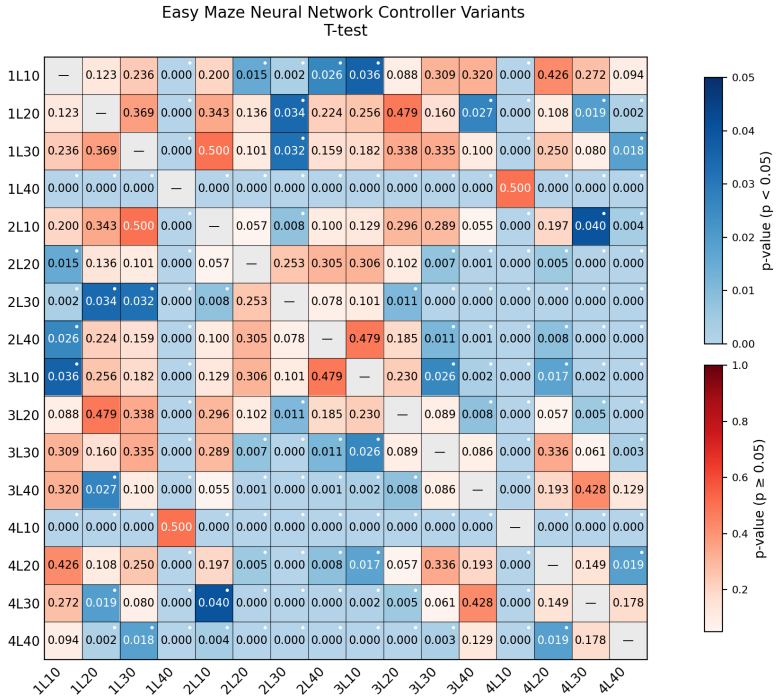


Fig. 29: Easy maze: pairwise one-tailed t-test matrix.

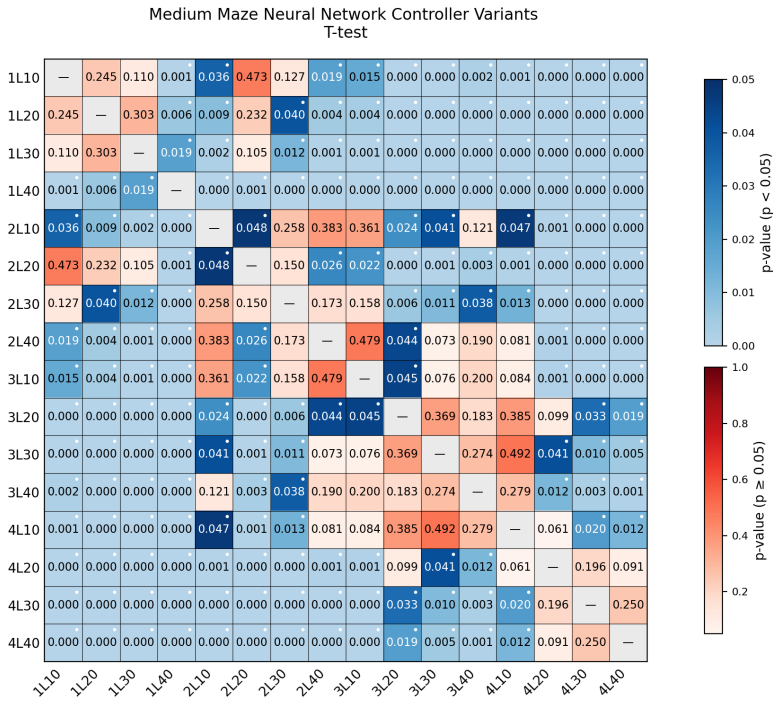


Fig. 30: Medium maze: pairwise one-tailed t-test matrix.

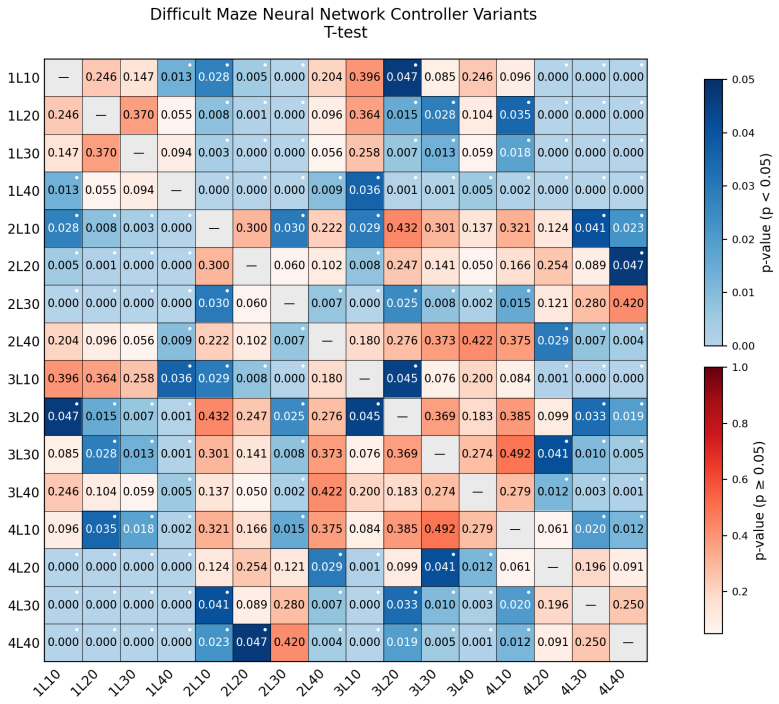


Fig. 31: Difficult maze: pairwise one-tailed t-test matrix.